

Sudden Cardiac Arrest and Death in Sports: An Updated Overview of Epidemiology, Etiologies, and Prevention Strategies, with Emphasis on Inherited Cardiomyopathies

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Abstract

Sudden cardiac arrest (SCA) and sudden cardiac death (SCD) in sports, although rare, remain pressing concerns with rising global participation in athletics. Recent epidemiological analyses indicate that incidence of SCA/SCD in athletes is low and has remained relatively stable over the past two decades. However, outcomes have improved as survival rates increase, reflecting advances in emergency response. Key risk factors are well-established, such as male athletes face a markedly higher risk than females, older master athletes have higher SCD risk rates than those under 35 years (largely due to acquired coronary artery disease), and certain high-intensity sports (e.g., basketball, American football, soccer, and endurance exercise) present higher SCA/SCD risk. In young competitive athletes, most SCDs are a consequence of undetected hereditary or congenital heart disorders. Predominant causes in this group include structural cardiomyopathies, such as hypertrophic cardiomyopathy and arrhythmogenic cardiomyopathy, and primary electrical diseases (channelopathies like long QT syndrome, Brugada syndrome, or catecholaminergic polymorphic ventricular tachycardia). Contrarily, occult atherosclerotic coronary disease becomes the leading cause of fatal events in master athletes. Multifaceted prevention strategies have expanded in recent years. Pre-participation cardiac screening programs aim to detect at-risk individuals and have successfully prevented SCD. Equally important, widespread implementation of emergency action plans, including on-site automated external defibrillators and cardiopulmonary resuscitation training, has improved resuscitation outcomes. Heightened public awareness, prompted by high-profile on-field cardiac arrests, has catalyzed policy changes that mandate better cardiac screening and emergency preparedness. Collectively, these efforts are shaping a safer sporting environment and reducing the SCA/SCD risk in athletes.

Keywords

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A significant increase in awareness of physical activity as a vital component of both mental and physical well-being has been observed in recent years. This growing recognition has contributed to a steady rise in individuals engaging in regular exercise and various sports worldwide. According to Statista,¹ an official source providing updated statistics on recreational sports participation, 21.1% of the U.S. population engaged in some form of daily sports or exercise in 2023, with a slightly higher participation rate among men (22.4%) than women (19.9%). Among outdoor activities, hiking was the most popular in the U.S. in 2023, followed by cycling, running/jogging, swimming, basketball, tennis, and soccer.

Specifically, cycling saw a substantial rise in participation, with approximately 56 million Americans engaging in recreational road cycling, mountain biking, or BMX in 2023, marking a 9.3% increase from 51.2 million in 2018. Conversely, running/jogging saw a slight decline, with 48.3 million participants in 2023 compared to 49.5 million in 2018 (−2.4%). Swimming remained a widely practiced sport, with 31.5 million participants in 2023, reflecting a 3% increase from 30.6 million in 2018. Basketball also experienced a notable rise in participation, growing from 24.2 million in 2018 to 29.7 million in 2023 (+22.7%). Tennis saw an even more significant surge, with 23.8 million players in 2023 compared to 17.8 million in 2018 (+33.7%). Soccer, a globally popular sport, also gained traction in the U.S., with participation rising from 11.4 million in 2018 to 14.1 million in 2023 (+23.7%).

A 2021 global Ipsos study analyzed sports and fitness trends among 21,503 online adults (ages 16–74) across 29 countries worldwide.² The global average for weekly exercise time was 6.1 hours, with the highest in the Netherlands (12.8 hours) and the lowest in Brazil (3 hours). Additionally, 14% of respondents reported not exercising at all, with Japan having the highest percentage (34%) and the Netherlands the lowest (4%). The five most practiced physical activities globally were general fitness (20%), running (19%), cycling (13%), soccer (10%), and swimming (9%). Notably, 58% of respondents desired to increase their sports participation, while only 6% intended to decrease it.

Regardless of these statistics, it is well-established that engaging in sports offers many health benefits, including improved cardiovascular health, weight management, enhanced muscular strength, and reduced stress levels. These factors collectively decrease the risk of several diseases, such as cardiovascular conditions, diabetes, cancer, and osteoporosis, while also delaying physical and cognitive decline and retardation of brain with aging.³ Additionally, sports participation fosters social connections and instills discipline.

However, physical activity is not without risks. Sports-related injuries, ranging from minor sprains and muscle strains to fractures and concussions, are common.⁴ Repetitive movements in certain sports can lead to chronic conditions like tendonitis and joint disorders.⁵ The pressure to perform at high levels may negatively impact mental health, leading to stress, anxiety, burnout, and depression.⁶ In some cases, athletes resort to performance-enhancing drugs,

which pose severe health risks.⁷ Although rare, sudden cardiac events may also occur, often triggered by strenuous exertion and/or undiagnosed heart conditions.

Recent Epidemiology of Sudden Cardiac Death in Athletes

To assess the incidence of sudden cardiac arrest (SCA) and sudden cardiac death (SCD) in athletes and military personnel, Lear et al⁸ conducted a systematic literature review and meta-analysis in 2021, incorporating data from 40 studies, 34 of which focused on athletes ranging from youth to professional levels. The reported incidence of SCD varied from 0.02 to 89.05 per 100,000 person-years, while SCA incidence ranged from 0.94 to 63.03 per 100,000 person-years. When restricting the analysis to population-based studies in athletes, the pooled SCD incidence was 0.98 (95% CI, 0.62–1.53) per 100,000 athlete-years, increasing to 1.91 (95% CI, 0.71–5.14) per 100,000 athlete-years in studies of competitive athletes. Due to a limited number of studies with a low risk of bias, a separate analysis for SCA prevalence in athletes was not conducted. However, two studies identified reported SCA incidences of 0.94⁹ and 1.58¹⁰ per 100,000 athlete-years.

D'Ascenzi et al¹¹ performed a meta-analysis in 2022 to investigate post-mortem etiologies of SCD in young individuals (≤ 35 years), comparing athletes and non-athletes. Their study incorporated data from 34 studies, encompassing 5,060 SCD cases (2,890 athletes and 2,170 non-athletes). Among athletes, the most common findings included a structurally normal heart (19% vs. 35.5% in non-athletes; $p=0.22$), hypertrophic cardiomyopathy (14.2% vs. 8.0%; $p=0.13$), idiopathic LVH (10.6% vs. 6.1%; $p=0.052$), anomalous coronary artery origins (9.7% vs. 4.4%; $p=0.14$), coronary artery disease (9.1% vs. 19.6%; $p<0.01$), myocarditis (6.1% vs. 8.9%; $p=0.21$), non-ischemic LV scar (5.1% vs. 1.1%; $p=0.01$), arrhythmogenic cardiomyopathy (4.7% vs. 11.5%; $p=0.03$), mitral valve prolapse (3.6% vs. 4.9%; $p=0.68$), valvular heart disease (3.3% vs. 2.0%; $p=0.07$), dilated cardiomyopathy (3.2% vs. 3.8%; $p=0.74$), aortic dissection (3.0% vs. 3.3%; $p=0.82$), and channelopathies (1.9% vs. 8.4%; $p=0.02$). Other causes accounted for 9.8% of cases in athletes and 13.5% in non-athletes ($p=0.35$).

Kim et al¹² conducted a literature review on SCA and SCD in competitive and recreational athletes in 2024, reporting that the incidence of SCA and SCD among young competitive athletes ranged from 1:2,400 to 1:417,000 person-years, decreasing to 1:15,000 and 1:64,000 person-years in more recent studies. Male athletes exhibited a 2- to 10-fold higher risk of SCA and SCD than females. The highest incidences were observed in basketball, football, cycling, and American football. Black athletes had up to a 5-fold increased risk compared to White athletes, regardless of age and sex. The study also highlighted that while epidemiological research has primarily focused on competitive athletes ≤ 35 years old, exercise-related SCA and SCD occur 7- to 10-fold more frequently in master athletes. This group estimated SCA and SCD incidences at 0.54 per 100,000 for half-marathon

Table 1 Leading causes of sudden cardiac death (SCD)/sudden cardiac arrest (SCA) in athletes

1. Hypertrophic cardiomyopathies
2. Idiopathic left ventricular (LV) hypertrophy
3. Nonischemic left ventricular (LV) fibrosis
4. Dilated cardiomyopathy
5. Anomalous coronary artery origins
6. Coronary artery disease
7. Myocarditis
8. Arrhythmogenic cardiomyopathies (e.g., Brugada syndrome, Wolff-Parkinson-White syndrome, long QT syndrome, and catecholaminergic polymorphic ventricular tachycardia)
9. Valvular heart disease (including mitral valve prolapse)
10. Aortic dissection
11. Channelopathies
12. Commotio cordis
13. Toxic cardiomyopathies (including sports doping)

runners, 0.39 per 100,000 for full-marathon runners, and 1.74 per 100,000 for triathletes. In athletes ≤ 35 years, autopsy-negative sudden unexplained death was the most common finding, followed by inherited and congenital cardiac conditions, including hypertrophic cardiomyopathy, coronary artery anomalies, mitral valve prolapse, arrhythmogenic cardiomyopathies, and genetic arrhythmia syndromes (e.g., Brugada syndrome, Wolff-Parkinson-White syndrome, long QT syndrome, and catecholaminergic polymorphic ventricular tachycardia). Acquired conditions such as myocarditis, commotio cordis, idiopathic myocardial fibrosis, and toxic myocarditis (including doping-related cases) were also implicated (► **Table 1**). In contrast, coronary artery disease was the predominant cause (between 30 and 80%) of SCA and SCD in master athletes.

A following UK-based registry analyzed 7,880 SCD cases between 1994 and 2022, including 848 athletes (11%; mean age: 33 ± 15 years).¹³ The majority (87%) of athlete-related SCDs occurred during exercise. Unlike prior meta-analyses, most deceased athletes were White (89%), followed by Black (6%) and Asian (5%). The most frequently involved sports were soccer (31%), running (20%), swimming (7%), and cycling (6%). The most common autopsy findings included structurally normal hearts suggestive of sudden arrhythmic death syndrome (SADS) (45%), myocardial diseases (32%, including LV fibrosis, hypertrophy, arrhythmogenic cardiomyopathy, and hypertrophic cardiomyopathy), coronary artery disease (7%), and coronary artery anomalies (3%).

Li et al¹⁴ conducted a systematic review and meta-analysis on sex differences in SCA/SCD incidence in competitive athletes in 2025. Analyzing data from 16 studies comprising 1,797 SCA/SCD cases, they found that 87.8% occurred in males, with an incidence of 1.42 (95% CI, 0.97–2.09) per 100,000 athlete-years in males and 0.32 (95% CI, 0.17–0.59) per 100,000 athlete-years in females (incidence rate ratio: 5.55). In males, hypertrophic cardiomyopathy (45.1%) was the leading cause of SCD, followed by congenital coronary

anomalies (16.0%), coronary artery disease (10.3%), myocarditis (6.2%), valvular heart disease (6.0%), and arrhythmogenic cardiomyopathies (5.3%). In females, congenital coronary anomalies were most common (33.4%), followed by hypertrophic cardiomyopathy (13.9%), arrhythmogenic cardiomyopathies (12.2%), myocarditis (10.4%), and valvular heart disease (8.7%).

Another interesting article was published in 2025 by Visanji et al,¹⁵ who conducted a retrospective cohort analysis to investigate the incidence and prognosis of sports-related SCA in Canadian patients aged 18 to 85 years. The study analyzed emergency medical services records from 2016 to 2020. Of the 18,769 recorded SCAs, 339 (1.8%) were sports-related, occurring predominantly in males (93.8%) with a mean age of 58 ± 14 years. The overall incidence of sports-related SCA was estimated at 1.2 per 100,000 person-years (95% CI: 1.1–1.4). Men exhibited a nearly 16-fold higher incidence than women (2.3 vs. 0.2 per 100,000 person-years). The most frequently involved sports included hockey, cycling, running, ball sports, gym workouts, and swimming. Notably, among patients with a documented final vital status, 52% survived hospital discharge.

Finally, in a large observational study by Kim et al,¹⁶ the authors investigated cardiac arrest incidence, causes, and outcomes among half-marathon and marathon runners between 2010 and 2023 in the United States. Drawing from over 29 million race finishers, the authors identified 176 cases of cardiac arrest, with an overall incidence of 0.54 per 100,000 participants, remarkably stable compared to the prior decade. However, the study reveals a significant improvement in the cardiac arrest mortality rate, which dropped from 71% in 2000 to 2009 to 34% in 2010 to 2023. Men and marathon participants were at notably higher risk than women and half-marathon runners. Coronary artery disease emerged as the most common underlying cause of cardiac arrest, displacing hypertrophic cardiomyopathy, which had previously been dominant. Survival was strongly associated with immediate

bystander cardiopulmonary resuscitation and access to defibrillation, highlighting the effectiveness of enhanced emergency preparedness in sports settings. The findings underscore the importance of primary cardiovascular screening, especially in older male runners, and suggest that medical readiness has significantly improved survival outcomes in long-distance races despite increased participation and the challenges of the COVID-19 era.

Causes of Sudden Cardiac Death

Age Group Differences

Age is a major determinant of SCA/SCD incidence in the athletic context.¹² Young competitive athletes (typically ≤ 35 years) have very low absolute rates of SCD, in the order of 0.47 to 1.21 per 100,000 persons.^{17–19} For example, in young athletes, an autopsy often finds no structural anomaly (autopsy-negative sudden arrhythmic death) or an inherited heart condition.²⁰ In Minnesota high school athletes, aged between 12 and 19 years, an SCD incidence of 0.24 per 100,000 athlete-years was reported.²¹ On the other hand, an annual incidence of SCD of 1:43,770 was reported in NCAA athletes (age 17–23 years).²²

In contrast, master athletes (older adults >35 years) experience exercise-related cardiac events several-fold more often, i.e., 6.64 per 100,000 person-years.^{17–19,23} Thus, SCA/SCD during exercise occurs more frequently in middle-aged and senior athletes than in athletes under the age of 35.²³ This is mainly due to the higher prevalence of acquired heart disease (especially coronary artery disease) in older individuals. In fact, in master athletes, coronary artery disease is the most likely entity to cause sports-related SCA/SCD.²³ Recent race registry data from 2010 to 2023 show that among endurance sport participants, coronary artery disease has surpassed hypertrophic cardiomyopathy as the leading cause of cardiac arrest in master athletes.¹⁶ Thus, while public attention often focuses on the tragic deaths of young athletes, the absolute burden of exercise-related SCA is higher in older adults, simply because far more people in their 40s, 50s, and 60s experience cardiac events during exercise (often unmasking silent coronary artery disease).¹²

It is important to note that the number of master athletes has increased over the past two decades, with more middle-aged adults participating in vigorous exercise and endurance sports, which may lead to a higher absolute count of sports-related SCAs in that demographic. However, parallel improvements in preventive cardiac screening (for high-risk individuals) and acute response (automated external defibrillators availability) at community sports events have likely mitigated the impact on SCD incidence. Many older athletes now undergo stress testing or coronary assessments, and sports like marathons often have robust medical coverage, which improves survival if an arrest occurs.

Sex Differences

Sex (biological gender) is one of the strongest predictors of SCA/SCD risk in sports. Numerous studies over the last 20 years have shown that males account for the vast majority

of sudden cardiac events in athletes, typically 80 to 95% of cases. Notably, even among females, the incidence of sports-related SCD remains very low (approximately 0.3–0.6 per 100,000 in most studies).¹⁴ A meta-analysis in 2025 found that 87.8% of athlete SCA/SCD cases were male.¹⁴ In high school and college cohorts, male athletes have an incidence of SCD that is anywhere from 2-fold to >10 -fold higher than that of female athletes.²⁴ The 2025 sex-specific review quantified this disparity: 1.42 per 100,000 in males vs. 0.32 per 100,000 in females, approximately $5.5\times$ difference in incidence.¹⁴ Similarly, in Japanese school sports, fatalities among boys far exceeded girls ($7.5\times$ higher rate).²⁵

Several factors contribute to this gap. Male athletes are more likely to have underlying congenital heart diseases that cause SCD (e.g., hypertrophic cardiomyopathy is more common and more severe in young men).²⁶ Males also participate in higher numbers in sports with greater cardiac demands, and historically had higher participation in sports overall, though female sports participation is steadily rising.^{27,28} Over the past 20 years, as female sports participation has increased, there has not been a disproportionate rise in female SCD; in fact, female athletes often undergo similar screening and benefit from the protective effect of a generally lower prevalence of dangerous cardiac anomalies.²⁹ Additionally, some conditions, like coronary artery disease (in master athletes) and myocarditis, are more prevalent or more lethal in men.^{30–32}

The sex disparity highlights the need for targeted prevention: screening algorithms and emergency preparedness often prioritize male athletes (since that is where most events occur), but female athletes should not be neglected. There have been cases of SCD in female athletes (often due to undiagnosed arrhythmias or congenital anomalies), and ensuring cardiac screening and automated external defibrillators access for women's sports is also important, even if the incidence is lower.

Sport-specific Differences

The type of sport and level of exertion play a significant role in modulating SCA/SCD risk. High-intensity team and endurance sports remain prominent contexts for SCA/SCD in athletes. In the U.S., basketball and American football continue to show the highest incidence among young competitive athletes.³³ In Europe (where soccer dominates participation), soccer is the most common sport in SCD cases.³³ Endurance sports like running, cycling, and swimming are consistently among the top activities associated with SCA/SCD across regions.^{26,34} A new Canadian registry (2016–2020) highlighted that even recreational sports can trigger SCA.¹⁵ For example, hockey, cycling, and running were frequent triggers in adult sports-related arrests, with an overall incidence of approximately 1.2 per 100,000 person-years.¹⁵ That study's cohort had a mean age of 58, reflecting the inclusion of many master athletes, and demonstrated that survival to hospital discharge reached approximately 52% in sports-related SCA, an encouraging outcome likely attributable to prompt bystander cardiopulmonary resuscitation and defibrillation. In the massive

analysis of over 29 million U.S. marathon and half-marathon finishers (2000–2023), the authors found that race-related cardiac arrest remains very low (approximately 0.5 per 100,000 participants) and remarkably stable over time.¹⁶ Crucially, however, the case-fatality rate in these events has fallen dramatically. This reflects improved emergency response (widespread automated external defibrillators availability, better training) and yields a much higher survival rate in recent years. In summary, new epidemiological evidence affirms known risk patterns (male sex, older age, certain sports) while also highlighting improving survival trends. It also suggests that while the absolute incidence of SCA/SCD in athletes remains low and relatively stable,³⁵ heightened awareness and preparedness make sports safer than ever when these rare events occur.

Most Frequent Genetic Causes of Sudden Cardiac Arrest/Death in Athletes

It is now well established that a broad spectrum of inherited cardiac disorders confers a significant risk of SCA and SCD. Over the past three decades, the introduction of next-generation sequencing (NGS) has markedly advanced the field of cardiovascular genetics, fostered the identification of many disease-associated genes, and shown that these disorders are genetically heterogeneous, often involving allelic variants. As previously discussed, the primary cause of SCA and SCD is coronary artery disease in adults over the age of 45. Within the pediatric population, as well as in young adults and athletes, SCA and SCD are instead more commonly attributed to inherited cardiac disorders, which can typically be categorized into two major classes. The first includes cardiomyopathies, in which the arrhythmogenic substrate primarily arises from structural abnormalities of the myocardium.³⁶ In contrast, the second class comprises channelopathies, where the heart is structurally normal, but arrhythmias result from dysfunctions in its electrical activity,³⁷ as summarized in ►Table 2.

Cardiomyopathies

Cardiomyopathies include dilated cardiomyopathy, hypertrophic cardiomyopathy, restrictive cardiomyopathy, and arrhythmogenic cardiomyopathy. These inherited disorders are traditionally classified based on morphological and functional characteristics of the heart. They have been classically regarded as Mendelian conditions, showing familial inheritance across generations and typically involving a single gene contributing to disease risk. However, most cardiomyopathies exhibit incomplete penetrance and variable expressivity, challenging genotype–phenotype correlations, even within the same family. The majority of genes identified in association with cardiomyopathies so far encode desmosomal, sarcomeric, cytoskeletal, or nuclear envelope proteins.³⁸

Dilated cardiomyopathy is clinically characterized by left ventricular dilation, myocardial fibrosis, and systolic dysfunction. SCA and SCD typically occur in the context of

progressive heart failure. A familial form is identified in approximately 20 to 35% of dilated cardiomyopathy cases. The condition is predominantly inherited in an autosomal dominant manner, although autosomal recessive, mitochondrial, and X-linked forms have also been reported.³⁹

Dilated cardiomyopathy is a clinically and genetically heterogeneous condition, with over 30 disease-associated genes identified to date. These genes encode a wide array of proteins involved in various cellular functions; however, four genes—titin (*TTN*), lamin A/C (*LMNA*), β -myosin heavy chain (*MYH7*), and cardiac troponin T (*TNNT2*)—account for the majority of genotype-positive dilated cardiomyopathy cases.⁴⁰ Dilated cardiomyopathy may present with additional clinical features such as arrhythmias and conduction abnormalities. These more complex phenotypes are most commonly associated with *LMNA*, *DES*, or *SCN5A* mutations. Notably, the identification of *LMNA* mutations in young individuals and athletes enables early surveillance for pre-clinical signs of conduction system disease. Timely intervention with pacemaker implantation or an implantable cardioverter-defibrillator is critical in these patients, given their elevated risk of malignant arrhythmias and subsequent SCA or SCD. In some cases, dilated cardiomyopathy occurs as part of multisystem syndromic disorders, including Duchenne and Becker muscular dystrophies, Emery-Dreifuss muscular dystrophy, Limb-Girdle muscular dystrophy, Barth syndrome, and myofibrillar myopathy.

Hypertrophic cardiomyopathy is characterized by myocyte disarray, myocardial fibrosis, and unexplained ventricular hypertrophy, typically with prominent involvement of the interventricular septum. This disorder is most commonly inherited as an autosomal dominant trait and is primarily associated with gene mutations encoding sarcomere components. Mutations in sarcomeric genes are identified in 50 to 60% of hypertrophic cardiomyopathy cases, with *MYBPC3* and *MYH7* being the most frequently implicated. However, recent studies have revealed that mutations in several genes encoding nonsarcomeric proteins also contribute to hypertrophic cardiomyopathy. These include Z-disk proteins such as *ACTN2* and *MYOZ2*, as well as intracellular calcium modulators like *JPH2*. Several distinct syndromes present with LV hypertrophy (LVH) similar to hypertrophic cardiomyopathy but differ in clinical presentation, inheritance patterns, and pathophysiology. These include Pompe disease (*GAA*), Danon disease (*LAMP2*), LVH with Wolff-Parkinson-White syndrome (*PRKAG2*), Fabry disease (*GLA*), and familial amyloidosis (*TTR*). These conditions share the feature of LVH but are distinguished by extracardiac manifestations and differences in genetic inheritance and disease mechanisms.

Arrhythmogenic cardiomyopathy is a progressive cardiac disorder characterized by structural alterations such as cardiomyocyte loss and fatty or fibro-fatty replacement of the myocardium. Arrhythmogenic cardiomyopathy is a clinically heterogeneous condition that is associated with systolic dysfunction, myocardial dilation, heart failure, and increased risk of SCA and SCD in young individuals. Historically, arrhythmogenic cardiomyopathy was thought to affect the right ventricle predominantly; however, recent studies

Table 2 Rare inherited cardiac disorders leading to SCA and/or SCD

	Disease	Clinical features	Inheritance	Major genes	Syndromic forms
Cardiomyopathies	DCM	Left ventricular dilatation, fibrosis and systolic dysfunction and SCA and SCD	AD AR XL Mt	Sarcomeric, cytoskeletal and nuclear envelope proteins (i.e., <i>TTN</i> , <i>LMNA</i> , <i>MYH7</i> , <i>TNNI2</i> , <i>LMNA</i> , <i>DES</i> , <i>SCN5A</i>)	Duchenne or Becker muscular dystrophy, Emery–Dreifuss muscular dystrophy, limb–girdle muscular dystrophy, Barth syndrome, myofibrillar myopathy
	HCM	Myocyte disarray and fibrosis and unexplained ventricular hypertrophy often with the involvement of the interventricular septum and SCA and SCD	AD AR	Sarcomeric, cytoskeletal and nuclear envelope proteins (i.e., <i>MYBPC</i> , <i>MYH7</i> , <i>ACTN2</i> , <i>MYOZ2</i> , <i>JPH2</i>)	Pompe disease; Danon disease; ventricular hypertrophy with Wolff–Parkinson–White syndrome; Fabry disease and familial amyloidosis
	ACM	Progressive cardiac disease leading to structural changes involving cardiomyocyte loss and fatty or fibro-fatty replacement of the myocardium and SCA and SCD	AD	Desmosomal genes (i.e., <i>PKP2</i> , <i>DSG2</i> , <i>SDC2</i> , <i>DSP</i> , <i>JUP</i> , <i>DES</i> , <i>TMEM43</i> , <i>PLN</i>)	Cardio-cutaneous syndromes: Naxos disease, Carvajal syndrome
	RCM	Diastolic dysfunction due to abnormal muscle relaxation and myocardial stiffness resulting in restrictive filling of the ventricles and SCA and SCD	AD	<i>cTNTT</i> , <i>cTNNI</i> , <i>MyBP-C</i> , <i>MYH7</i> , <i>MYL2</i> , <i>MYL3</i> , <i>DES</i> , <i>MYPN</i> , <i>TTN</i> , <i>BAG3</i> , <i>DCBLD2</i> , <i>LMNA</i> , <i>FLNC</i>	–
Channelopathies	LQTS	Prolongation of cardiac repolarization leading to ventricular arrhythmias; stress-induced syncope episodes and SCA and SCD	AD	Genes encoding for subunits of ion channels and in genes encoding ion channels' regulatory proteins (i.e., <i>KCNQ1</i> , <i>KCNH2</i> , <i>SCN5A</i>)	–
	SQTS	Characterized by a QT interval shorter than the normal, ventricular and atrial arrhythmias	AD	Genes encoding for subunits of ion channels and in genes encoding ion channels' regulatory proteins (i.e., <i>KCNQ1</i> , <i>KCNH2</i> , <i>SCN5A</i>)	–
	BrS	Peculiar ECG pattern characterized by ST-segment elevation of the right precordial leads with or without right bundle branch block in the absence of structural heart disease	AD	<i>SCN5A</i> , <i>KCNE3</i> , <i>CACNA1c</i> , <i>CACNB2</i>	–
	CPVT	Adrenergically mediated life-threatening arrhythmias causing syncope, SCA and SCD in pediatric age	AD AR	Genes encoding for calcium channels (i.e., <i>RyR2</i> , <i>CASQ2</i>)	–

Abbreviations: ACM, arrhythmogenic cardiomyopathy; AD, autosomal dominant; AR, autosomal recessive; BS, Brugada syndrome; CPVT, catecholaminergic polymorphic ventricular tachycardia; DCM, dilated cardiomyopathy; HCM, hypertrophic cardiomyopathy; LQTS, long QT syndrome; Mt, mitochondrial inheritance; RCM, restrictive cardiomyopathy, SCA, sudden cardiac arrest; SCD, sudden cardiac death; SQTS, short QT syndrome; XL, X-linked inheritance.

have identified biventricular and left ventricle variants of the condition.⁴¹ Arrhythmogenic cardiomyopathy is inherited in an autosomal dominant pattern, with estimates suggesting that 30 to 50% of patients harbor a pathogenic or likely pathogenic variant in a gene encoding a desmosomal protein, including plakophilin (*PKP2*), desmoplakin (*DSP*), plakoglobin (*JUP*), desmoglein-2 (*DSG2*), and desmocollin-2 (*DSC2*). Mutations in nondesmosomal genes such as *DES*, *TMEM43*, and *PLN* have also been associated with this condition. However, it remains unclear whether these genes are directly causing arrhythmogenic cardiomyopathy or represent phenocopies of the disease. Arrhythmogenic cardiomyopathy may also present with clinical variants that involve systemic manifestations, including the recessively inherited cardio-cutaneous syndromes Naxos disease and Carvajal syndrome, which result from recessive mutations in the *JUP* and *DSP* genes. Restrictive cardiomyopathy is clinically characterized by diastolic dysfunction resulting from impaired muscle relaxation and increased myocardial stiffness, which leads to restrictive filling of the ventricles. Restrictive cardiomyopathy is the rarest form of cardiomyopathy.^{42f} Approximately 30% of restrictive cardiomyopathy cases are familial, with several genes implicated in this condition, including *TNNT2*, *TNNI3*, *MyBP-C*, *MYH7*, *MYL2*, *MYL3*, *DES*, *MYPN*, *TTN*, *BAG3*, *DCBLD2*, *LNMA*, and *FLNC*.⁴²

With the advent of NGS, which offers exceptional capabilities for high-throughput and cost-effective DNA analysis, it has become possible to perform genetic screening in large patient cohorts. This has led to the identification of multiple mutations in dilated cardiomyopathy, hypertrophic cardiomyopathy, and arrhythmogenic cardiomyopathy, including homozygous and compound heterozygous mutations within the same gene and double or multiple heterozygous mutations across different genes. These mutations are observed in 5 to 38% of patients with sporadic or familial forms of cardiomyopathy.^{43–45} These findings support the hypothesis that compound or double pathogenic variants may exert a gene-dose effect, potentially leading to more severe manifestations of the associated phenotype. Although reports on this subject remain limited, the available evidence suggests that patients with multiple mutations tend to have a more severe clinical course, including earlier disease onset, more rapid symptom progression, and earlier onset of significant complications. This suggests that the identification of multiple disease-causing mutations in affected subjects may serve as an important marker for risk stratification. This is particularly relevant for young athletes, who may require more aggressive treatment and earlier implementation of preventive strategies.⁴⁶

Special caution is required regarding the correlation between exercise or sports activity and cardiomyopathies, particularly in relation to SCA and SCD. There is now a growing trend toward sedentary lifestyles, which, in turn, has contributed to an increase in obesity and cardiovascular diseases. Consequently, recommendations for regular physical activity have become essential. It is well established that exercise serves as a critical component of therapy for many cardiovascular diseases and can reduce mortality rates. However, the literature suggests that physical activity may

act as a trigger for SCA or SCD in individuals with cardiomyopathies.

Due to the detrimental effects of a sedentary lifestyle, recent guidelines recommend low- to moderate-intensity physical activity, even for patients with cardiomyopathies. The latest updates to the European Society of Cardiology (ESC) guidelines on sports participation in patients with cardiovascular disease incorporate specific recommendations based on disease type and individual risk stratification.⁴⁷

In recent years, an increasing number of studies have shown that low or moderate physical activity does not accelerate disease progression in patients with inherited cardiomyopathies and does not significantly impact disease outcomes, except for arrhythmogenic cardiomyopathy. In fact, several studies have demonstrated that even moderate physical activity can be a modulating factor in arrhythmogenic cardiomyopathy, potentially promoting disease penetrance in genetically positive carriers, exacerbating ventricular dysfunction, and triggering ventricular arrhythmias, which may lead to SCA or SCD.⁴⁸ It is hence widely agreed that arrhythmogenic cardiomyopathy patients, particularly those with positive genetic findings, should be discouraged from participating in competitive sports and high-intensity leisure-time exercise. However, arrhythmogenic cardiomyopathy patients, especially young individuals with mild disease, should not be deprived of the benefits of personalized, low-intensity exercise, which experienced physicians and exercise physiologists can prescribe.^{49–51}

In conclusion, patients with cardiomyopathy, particularly those who are genotype-positive, are advised to avoid high-intensity exercise and competitive sports due to the documented increased risk of SCA and SCD. Nevertheless, they are encouraged to engage in low-intensity physical activity, as it is an important therapeutic strategy to reduce mortality.⁴⁷

Channelopathies

An increasing number of electrical heart disorders have been described in recent decades, particularly in young individuals, including athletes. These conditions are linked to various heritable disorders, such as long QT syndrome, short QT syndrome, Brugada syndrome, and catecholaminergic polymorphic ventricular tachycardia.³⁷ Long QT syndrome is characterized by a prolongation of cardiac repolarization, which can lead to ventricular arrhythmias. Symptomatic individuals often present with stress-induced syncopal episodes and are at risk of SCA or SCD. Following the discovery of familial recurrence of long QT syndrome, the first three genes identified were *KCNQ1*, which encodes the potassium current channel; *KCNH2*, encoding the repolarizing potassium current channel; and *SCN5A*, encoding the alpha-subunit of the sodium channel responsible for depolarizing sodium current. Pathogenic variants in additional genes, which encode for subunits of various ion channels and regulatory proteins, have been identified recently in long QT syndrome patients; so nearly 15 genes have been currently associated with long QT syndrome.

Short QT syndrome, in contrast, is characterized by a shortened QT interval, along with ventricular and atrial

arrhythmias and an increased risk of SCA and SCD. Although a limited number of short QT syndrome cases have been reported, SCA or SCD often presents as the first manifestation of the disorder. Short and long QT syndromes are known to be allelic disorders, and most of the genes implicated in short QT syndrome encode subunits of ion channels or their regulatory proteins.

Brugada syndrome is a channelopathy marked by a distinct electrocardiogram (ECG) pattern, which shows ST-segment elevation in the right precordial leads with or without the right bundle branch block in the absence of structural heart disease. The most common clinical manifestations of Brugada syndrome are syncope or SCD, typically triggered by ventricular tachyarrhythmias occurring during sleep or at rest. Brugada syndrome is inherited in an autosomal dominant manner, although the majority of clinically diagnosed patients are male. The role of gender in modulating the disease manifestation remains unclear. The first gene associated with Brugada syndrome was *SCN5A*, which encodes the cardiac sodium channel, the same gene linked to *LQT3*. Although most currently genotyped Brugada syndrome patients carry pathogenic variants in *SCN5A*, recent studies have reported additional pathogenic variants in other genes, such as *KCNE3*, *CACNA1c*, and *CACNB2*.⁵²

Catecholaminergic polymorphic ventricular tachycardia presents with adrenergically mediated life-threatening arrhythmias that cause syncope, SCA, and/or SCD, typically beginning during childhood. The surface ECG is often normal, and diagnosis is mainly based on symptoms and detection of stress-induced arrhythmias during an exercise stress test or Holter monitoring. Catecholaminergic polymorphic ventricular tachycardia is one of the most lethal channelopathies, with a natural history showing that up to 30% of affected individuals experience SCA or SCD before the age of 40 in the absence of antiadrenergic therapy, particularly among young athletes.⁵³ The identification of genetic mutations in *RyR2* and *CASQ2* has significantly advanced our understanding of the molecular basis of these arrhythmias; pathogenic variants in these genes lead to increased calcium release from the sarcoplasmic reticulum, promoting the onset of triggered arrhythmias. A major advancement from genetic studies in channelopathies has been the recognition of the role of these variants in determining the risk of SCA and SCD. Various studies have shown that a family history of SCA or SCD is a significant risk factor, suggesting that genetic factors influence this predisposition.⁵⁴ For channelopathies, especially catecholaminergic polymorphic ventricular tachycardia, genetic testing has proven to be beneficial in identifying patients at high risk of SCA and SCD, particularly athletes. Variants in *RYR2*, although electrically silent, can be detected through genetic testing, providing significant advantages in the diagnosis and prognosis of disease.

Advances in Genetic Screening and Inherited Cardiovascular Risk

Breakthroughs in genetic screening over the past 2 years are influencing how we identify and manage athletes at risk for

sudden death.⁵⁵ Research in 2024 to 2025 has especially focused on inherited cardiomyopathies and channelopathies, the conditions often behind SCA in apparently healthy young athletes.⁵⁶ One pivotal development is the growing evidence that genotype-positive athletes without overt phenotype (“G+/P–” athletes) may safely continue sports under proper management.⁵⁷ Historically, a positive genetic test for a pathogenic variant (for example, in a hypertrophic cardiomyopathy or long QT syndrome gene) often meant automatic disqualification from competitive sports. However, a 2024 study followed athletes with disease-causing genetic variants but no or minimal clinical manifestations.⁵⁸ With individualized therapy (e.g., beta-blockers for long QT syndrome, implantable cardioverter defibrillators for the highest-risk cases, etc.), these athletes could return to play.⁵⁸ Over a combined 1,300 athlete-years of follow-up, none suffered an SCD or life-threatening cardiac event.⁵⁸ Notably, nearly 20% of these athletes had previously been unfairly disqualified elsewhere purely based on genotype.⁵⁸ This finding is a game-changer, suggesting that blanket restrictions based solely on genetic test results are overly conservative and that many gene-positive athletes can compete safely with expert oversight and shared decision-making. This aligns with a broader paradigm shift toward personalized risk stratification.

In terms of diagnostics, genetic testing technology has continued to advance. Panels for cardiomyopathy and channelopathy genes are now widely available and increasingly affordable.⁵⁹ The yield of genetic testing in unexplained young sudden death (so-called “molecular autopsy”) and in athletes with concerning family histories has been high.⁶⁰ For example, pathogenic variants can be identified in up to approximately 30 to 40% of autopsy-negative SCD cases.⁶¹ Recent publications underscore the importance of cascade screening in families of athletes who have died or who are diagnosed with inherited conditions.⁶² For clinicians, an emerging challenge is interpreting variants of uncertain significance and knowing when a positive genotype truly warrants sports restriction. Here, new data provide guidance. For instance, some channelopathies once thought incompatible with sports are being re-evaluated. In particular, catecholaminergic polymorphic ventricular tachycardia had long carried a universal recommendation against competitive sports.⁶³ But experts now report that with appropriate therapy (e.g., beta-blockers, implantable cardioverter-defibrillator) and close monitoring, select patients with catecholaminergic polymorphic ventricular tachycardia have successfully returned to sport.²³ This nuanced approach is echoed for conditions like long QT syndrome, where moderate QT prolongation can often be managed such that an athlete may continue training with precautions. In contrast, more malignant cases still warrant restriction.²³

Another area of progress is genetic risk markers for cardiomyopathy.⁶⁴ Although no brand-new “SCD genes” have been discovered in the past 2 years, we have a better grasp of the polymorphisms that modulate risk. For example, research in dilated cardiomyopathy and arrhythmogenic cardiomyopathy has identified how certain genetic variants

(like specific *LMNA* or *DSP* mutations) confer especially high arrhythmic risk.⁶⁵ Such insights can guide preventive implantable cardioverter-defibrillator placement or closer follow-up in athletic individuals who carry these mutations. There is also increasing interest in polygenic risk scores for conditions like hypertrophic cardiomyopathy.⁶⁶ Although still in early stages, the idea is to aggregate multiple genetic factors to stratify which athletes might have severe disease or SCD risk.

In summary, the 2024 to 2025 period has seen a continued shift from a one-size-fits-all genetic exclusion toward individualized management of genetic heart diseases in athletes. The key breakthroughs include evidence supporting safe sports participation for genotype-positive/phenotype-negative athletes under expert care, and updated guidelines reflecting this leniency. Genetics is no longer just for explaining SCD after the fact but a tool for preventing sudden death by identifying at-risk athletes early and tailoring their care (from lifestyle modifications to prophylactic interventions). Ongoing registries and studies will provide invaluable data on how these athletes fare in the long term with continued sports participation.

Major Influences on Incidence Trends

Pre-participation Screening

Widespread screening of athletes for heart disease has been a cornerstone of prevention strategy.⁶⁷ As noted, Italy's mandatory ECG screening program dramatically lowered the incidence of SCD among young athletes.⁶⁷ Following evidence of its success, many European countries adopted similar requirements by the 2010s.⁶⁷ In the U.S., formal ECG screening is not universally mandated, but there has been increasing adoption at the collegiate and even high school level.⁶⁸ A 2018 "Power 5" National Collegiate Athletic Association (NCAA) conference survey showed that 58% of universities added ECGs to the standard history/physical examination screening.⁶⁹ Expert consensus (including the American Heart Association 14-point evaluation) has gradually shifted toward more rigorous cardiac screening of athletes, especially males in high-risk sports.^{70,71} The impact of screening is evidenced by identifying conditions like hypertrophic cardiomyopathy, Wolff-Parkinson-White syndrome, and arrhythmogenic cardiomyopathy in athletes who get treatment or are restricted from play.²³ These interventions undoubtedly prevented some SCD events, though quantifying this prevention is challenging. Targeted genetic testing for known pathogenic mutations (for cardiomyopathies, channelopathies) is becoming more feasible and may further help identify at-risk young athletes.

Emergency Action Plans and Automated External Defibrillators Deployment

A major change since the mid-2000s is the vastly improved preparedness for SCA/SCD at athletic sports. Schools, gyms, and sports leagues increasingly have automated external defibrillators on-site and staff trained in cardiopulmonary resuscitation. This has improved survival rates and thereby reduced the incidence of death, even if arrests still occur.

In the U.S., the NCAA observed a 70% decline in SCD incidence, which is partly attributed to better resuscitation.²⁹ Therefore, many athletes who would have died 20 years ago are now successfully revived and thus not counted as SCD. Sports medicine programs instituted emergency action plans requiring rapid cardiopulmonary resuscitation and defibrillation. For example, high schools participating in an automated external defibrillator program reported 71% survival of student-athlete cardiac arrests in one series.⁷² Professional leagues and large race organizers have also placed medical teams and defibrillators strategically (the quick response to on-field collapses in several high-profile cases in the 2010s and early 2020s demonstrated the life-saving potential). This trend does not necessarily reduce the incidence of SCA but lowers SCD (fatal outcomes), which is reflected in epidemiology as a stable or declining SCD rate. This is a direct result of better emergency response.

COVID-19 Pandemic

The Coronavirus Disease 2019 (COVID-19) pandemic era (especially between the years 2020 and 2021) raised concerns about increased cardiac events in athletes due to myocarditis from Severe Acute Respiratory Distress Syndrome Coronavirus 2 (SARS-CoV-2) infection. Sports cardiology experts initially feared a surge in SCA as athletes returned to play. Fortunately, studies have not found a significant increase in exercise-related SCA/SCD during the pandemic. Astley et al³⁵ analyzed the frequency of SCA/SCD among young competitive athletes before and during the COVID-19 pandemic using a 6-year dataset (2017–2022) from the National Center for Catastrophic Sports Injury Research (NCCSIR). They identified 387 cases (mean age: 16.5 ± 2 years; 86.3% males). The incidence of SCA/SCD was not significantly different pre-pandemic versus during the pandemic (203 vs. 184 cases; $p = 0.33$). The proportion of exercise-related SCA/SCD cases remained unchanged during the COVID-19 pandemic (69.5%), indicating stability. Similarly, in the NCAA, only one SCD was linked to myocarditis in 2020 to 2022, and no SCDs were attributed directly to SARS-CoV-2 infection.⁷³ Many sports programs implemented cardiac MRI screening for athletes who had COVID-19, which may have helped exclude those with active myocarditis from play.

Additionally, organized sports were paused or reduced in many regions during the height of the pandemic (2020), which likely offset any infection-related risk by temporarily lowering exposure to intense exercise. By 2022, data suggested that the cumulative frequency of SCA/SCD remained essentially unchanged through the COVID period. However, the pandemic did spur increased cardiac screening protocols (e.g., clearance exams post-COVID) that might continue to benefit athlete safety.

Public Awareness and High-profile Cases and Influences on Policy (2024–2025)

In the past 2 years, several high-profile athletes' cardiac incidents have garnered global attention, spurring changes in policy and public awareness. One of the most impactful was the on-field cardiac arrest of National Football League (NFL) player Damar Hamlin in January 2023 (televised

nationally during a Monday Night Football game).⁷⁴ Immediate cardiopulmonary resuscitation and defibrillation saved Hamlin's life, and his miraculous recovery turned him into a vocal advocate for SCA preparedness. Hamlin partnered with major organizations to launch the "Smart Heart Sports Coalition," which has successfully advocated for legislation requiring automated external defibrillators and emergency response plans in youth sports. By mid-2024, multiple U.S. states (e.g., Ohio, New York, Texas, and others) had introduced or passed laws mandating automated external defibrillators availability in schools and sports, as well as annual training for coaches in recognizing and responding to SCA. For example, Ohio enacted a law in 2024 ensuring all schools have automated external defibrillators on site, and Washington State now requires automated external defibrillators in all gyms and fitness facilities. Recent athlete emergencies directly influenced these policy moves. The legislative discussions frequently cited Hamlin's case as evidence that preparedness saves lives. In addition, the collapse of Bronny James (a college basketball player and son of LeBron James) during a University of Southern California practice in July 2023 further underscored that SCA can strike elite young athletes.⁷⁵ Bronny James, like Hamlin, survived due to prompt defibrillation. His incident, occurring just months after Hamlin's, reinforced calls for widespread automated external defibrillator deployment and cardiac screening at the high school and collegiate level. Thanks to these well-publicized rescues, public awareness of SCA in youth athletes likely has never been higher than in 2024 to 2025.

Globally, other notable cases have continued to shape attitudes. The survival and return to professional play of Danish soccer player Christian Eriksen (who collapsed from SCA during a Euro 2021 match and now plays with an implanted defibrillator) provided a powerful success story that resonated into 2024.⁷⁶ It demonstrated that immediate cardiopulmonary resuscitation/automated external defibrillator use can be lifesaving, and with an implantable cardioverter-defibrillator for secondary prevention, athletes might even continue their career. This helped reduce stigma around devices like implantable cardioverter-defibrillators in athletes and has been cited in discussions by sports cardiologists advocating for case-by-case return-to-play decisions. High-profile tragedies and near-fatal incidents have significantly influenced clinical practices and safety protocols. For example, the young American basketball player Keyontae Johnson in 2020 (who collapsed on the court coming out of a timeout but eventually survived after intensive treatment), and others have driven home the need for proactive screening (including consideration of genetic testing or advanced imaging when warning signs like syncope or murmurs are present).

In 2024, the Cardiac Safety Research Consortium's Think Tank on SCD in the young convened experts to chart a "pathway to survival," inspired partly by these public cases.³⁵ The report emphasizes nationwide implementation of evidence-based emergency protocols in all sporting arenas. This includes having a clear emergency action plan, rapid access to automated external defibrillators, and regular rehearsal of drills, measures that were once left to individual institutions

but are increasingly standardized. Even sports organizations like the NCAA and professional leagues have heightened their cardiac emergency preparedness standards by learning from these incidents. For instance, the NCAA in 2023 to 2024 promoted that every team's athletic training staff review and practice their SCA response plan annually (a recommendation likely to become a requirement). The NFL and other leagues have also expanded the availability of emergency equipment and personnel at games since witnessing a high-profile cardiac arrest on the field.

Therefore, recent athlete cardiac events have accelerated improvements in policy and practice. In the late 2000s and 2010s, several U.S. states enacted laws (often named after student athletes who died) mandating better education on SCA for coaches, immediate removal from play for any athlete who faints, and automated external defibrillators availability at school sports. Hamlin's and Bronny James' cases in particular galvanized lawmakers and stakeholders to improve automated external defibrillator access in 2024. The conversation has shifted from "should we invest in rare emergencies?" to an understanding that every school and sporting venue must be prepared because seconds count when SCA strikes. As of 2025, more athletes than ever are competing under safer conditions, with trained responders and defibrillators at the ready, a direct positive outcome from difficult lessons learned.

Internationally, sports governing bodies (Fédération Internationale de Football Association (FIFA), International Olympic Committee (IOC), etc.) launched campaigns for heart health in sports. For instance, FIFA began a program to donate defibrillators to clubs worldwide after a series of on-field deaths. These efforts and media coverage of saves have made coaches, players, and spectators more vigilant. We now regularly see previously unnoticed warning signs (like syncope or palpitations in athletes) being taken seriously and investigated, which can lead to preventative treatment (e.g., an athlete getting an implantable cardioverter-defibrillator or medical therapy). All these factors contribute to the observed epidemiological trends: relatively stable incidence of SCA/SCD over 20 years but with improved outcomes. In short, the rate of these events has not dramatically risen despite more people engaging in sports, indicating success in prevention, and the fatality rate has decreased, indicating success in acute management.

Conclusion

The SCD in athletes remains a rare but devastating event that demands continued vigilance. Over the past two decades, research has solidified our understanding of key risk factors (including male sex, advancing age, and participation in high-intensity sports) that predispose athletes to SCA/SCD. At the same time, the incidence of sports-related SCA has held relatively steady, and fatality rates have significantly declined, despite an expanding athletic population.

Several studies have shown that pathogenic or likely pathogenic variants can be found in approximately 65% of patients with hypertrophic cardiomyopathy, 60% of patients

with arrhythmogenic cardiomyopathy, 75 to 80% of patients with long QT syndrome, 60% of patients with catecholaminergic polymorphic ventricular tachycardia, 20 to 30% of patients with Brugada syndrome, and up to 30% of patients with genetic cardiomyopathies.³⁷ Identifying these variants is crucial for patient management and significantly impacts the cascade screening of family members. It is also noteworthy that a family history of cardiomyopathy is an important consideration in the evaluation of young, asymptomatic athletes planning to participate in competitive sports, warranting comprehensive clinical assessment. Since SCA and SCD can occur in the early stages of inherited cardiomyopathies and channelopathies, often as the first manifestation of these disorders, particularly in young athletes, molecular genetic screening may be the only means for identifying young patients at risk of these fatal events. This has been tragically demonstrated in the cases of high-level athletes such as Puerta, Morosini, and Astori.

The decreasing trends in SCD reflect the positive impact of comprehensive prevention strategies and improved on-site cardiac emergency management. Mandatory or recommended pre-participation screening programs (ranging from focused cardiac questionnaires to ECG and imaging protocols) have enabled earlier detection of silent cardiomyopathies and electrical syndromes, allowing timely interventions or sports restrictions when necessary. Although inherited cardiac disorders remain a significant socioeconomic issue in industrialized countries, despite mandatory pre-participation screenings, gene panel screening has become more high-throughput and cost-effective with the advent of NGS. To this end, targeted screening for cardiomyopathies and channelopathies in selected young athletes could be an effective approach for identifying individuals at heightened risk for SCA and SCD.

This approach could eventually lead to the implementation of personalized medicine, the gold standard for managing rare genetic disorders. Advances in genetic testing and counseling now facilitate more nuanced risk stratification. Notably, a paradigm shift is underway: athletes who are genotype-positive for dangerous cardiac conditions but lack clinical disease are no longer automatically barred from competition. Emerging data suggest that many such individuals can safely continue to play under expert monitoring and therapy. Nonetheless, due to possible associations between certain cardiomyopathies and some substances potentially used by competitive athletes such as beta-blockers (affect heart rate and contractility, potentially masking symptoms or altering exercise tolerance), drugs that prolong QT interval (increase torsades de pointes risk), vasodilators and antihypertensives (can cause hypotension during exercise, potentially leading to syncope or ischemia in vulnerable hearts), stimulants such as caffeine and amphetamines (enhance adrenergic tone and myocardial oxygen demand, precipitating arrhythmias in susceptible individuals), anabolic steroids and performance-enhancing drugs (can lead to structural cardiac changes, arrhythmias, and increased risk of SCD) as well as other recreational drugs such as cocaine and cannabis (may trigger vasospasm, arrhythmias, and exacerbate underlying cardiac conditions), careful

medical supervision and avoidance of these high-risk substances would be essential in high-risk population. The extensive implementation of emergency action plans, such as easily accessible defibrillators and trained responders at sporting events, significantly enhances survival rates for cardiac arrests on the field. Increased public awareness and notable successful resuscitations have further driven policy reforms to make competitive sports safer.

Conflict of Interest

None declared.

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